Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

SPARKvue: Motion Sensors

Match Graph lab • PASCO wireless motion sensor

1. Open SPARKvue

Take out your **laptop**, open **Google Chrome**, and go to:

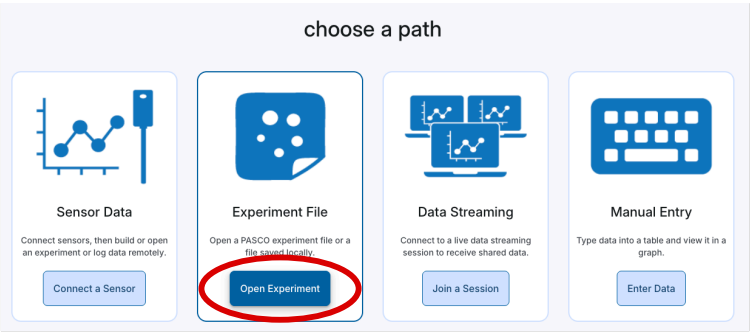
sparkvue.pasco.com

2. Download the lab file

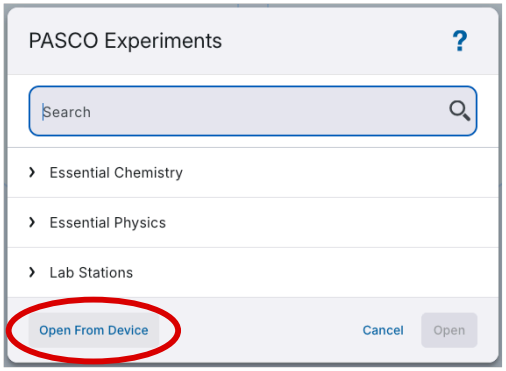
On your **class page**, click **Match Graph lab file**. It saves to your **Downloads** folder.

3. Open the lab file in SPARKvue

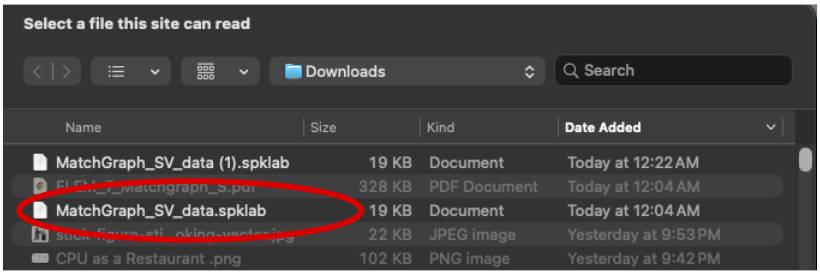
1. On the **choose a path** screen, under **Experiment File**, click **Open Experiment** (a)
2. In the **PASCO Experiments** box, click **Open From Device**, bottom left (b)
3. Open your **Downloads** folder, click **MatchGraph**, then click **Open** (c)



a. Open Experiment

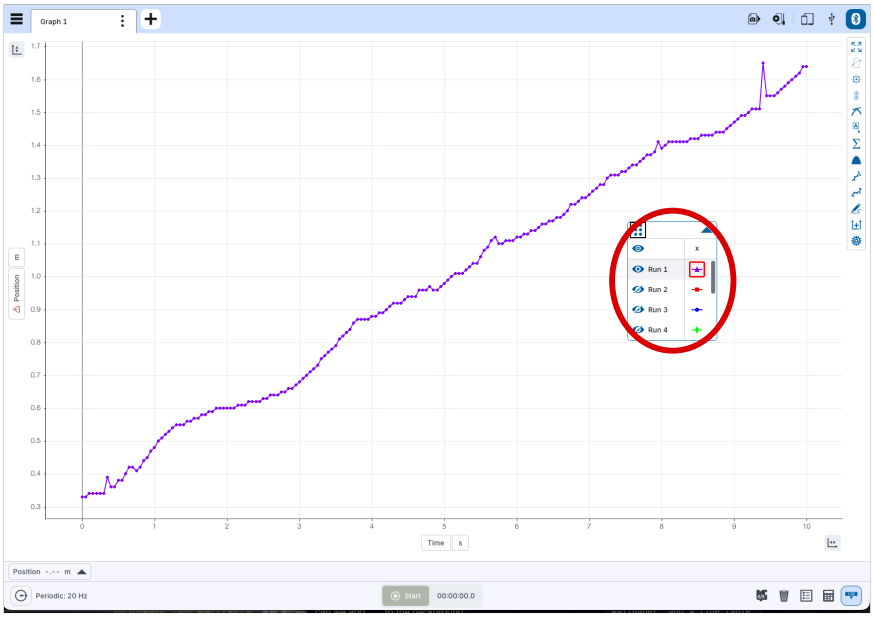


b. Open From Device



c. Downloads → MatchGraph → Open

Check: SPARKvue opens a graph like the one below. **Run 1** is showing; the other runs are hidden. **Runs 1 to 9** are the nine target graphs, drawn in *The nine runs*.



d. The lab file, open. The ringed box is the **legend**: click an **eye** to show or hide a run.

4. Connect the sensor

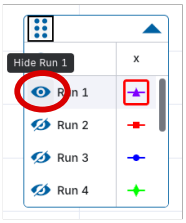
1. Put the sensor on a table, its **round screen at waist height**, facing **3–4 m** of clear floor.
2. Press the **power button**. The light **blinks red**.
3. In SPARKvue, click **Bluetooth** and pick the number printed on your sensor's label.

Check: the light now **blinks green**.

5. Match a run

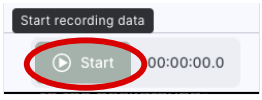
The sensor measures **your distance from its round screen**. Walk **away** and the line goes **up**; walk **towards** it and the line goes **down**. Stay at least **0.3 m** away.

1. **Show one run.** Have **only one of Runs 1–9 visible at a time**. Click the **eye** beside a run to **hide** or **show** it. Start with **Run 1** showing and every other run hidden.



Open eye: the run is **showing**.
Crossed-out eye: the run is **hidden**.
Here Run 1 is showing; Runs 2–4 are hidden.

2. **Stand** where the run's line starts.
3. **Walk it.** Your partner clicks **Start** (*Start recording data*), bottom of the screen; it records for **10 s**. Your walk appears in the legend as a new run.



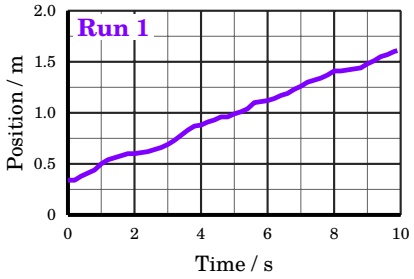
4. **Compare and retry** until your line follows the target run. Hide each old attempt with its eye, so the screen shows the target and your newest walk.
5. **Write** how you moved under that run in *The nine runs*, and tick *Matched?*

Check: everyone walks at least two runs. **Care:** keep the floor clear, and walk.

The nine runs

Words to use: stationary • constant speed • away from • towards • faster • slower • speeding up • slowing down • turn round

Runs 1–9 from PASCO's MatchGraph! lab for SPARKvue, redrawn from the lab file. Each run is drawn in its colour in the SPARKvue legend.



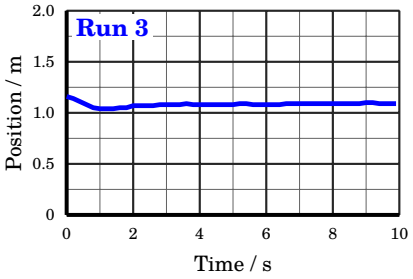
How I moved: **Matched?** ☐

Walked away at a slow, constant speed, from 0.3 m to 1.6 m.



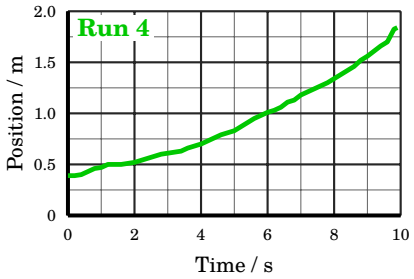
How I moved: **Matched?** ☐

Started 1.7 m away and walked towards, quickly at first, then slowing down. Finished 0.3 m away.



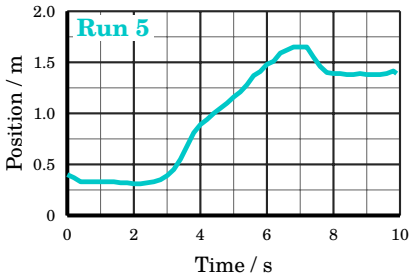
How I moved: **Matched?** ☐

Stood still (stationary) about 1.1 m from the sensor for the whole 10 s.



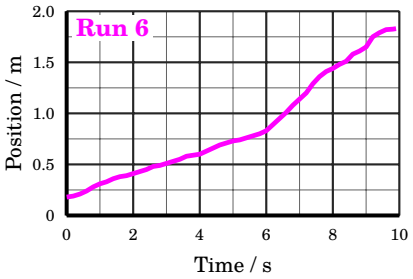
How I moved: **Matched?** ☐

Started 0.4 m away and walked away, slowly at first and speeding up, to 1.8 m.



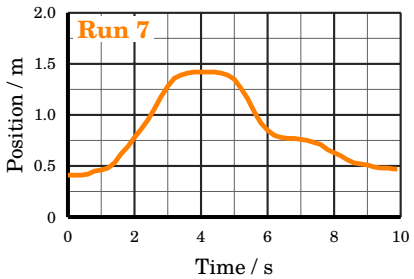
How I moved: **Matched?** ☐

Stood at 0.3 m for 3 s; walked away quickly to 1.65 m by 7 s; stepped back towards to 1.4 m; stood still.



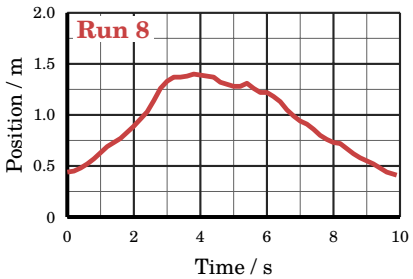
How I moved: **Matched?** ☐

Walked away slowly and steadily from 0.2 m to 0.8 m in 6 s, then walked away faster to 1.8 m.



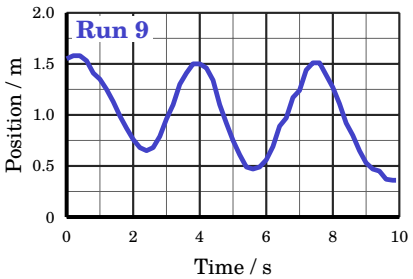
How I moved: **Matched?** ☐

Walked away, speeding up, to 1.4 m by 3.5 s; stood for about a second; walked back towards quickly, then more slowly, to 0.5 m.



How I moved: **Matched?** ☐

Walked away at a fairly steady speed to 1.4 m by 3.5 s, then walked back towards at a steady speed to 0.4 m.



How I moved: **Matched?** ☐

Back and forth: towards to 0.7 m, away to 1.5 m, towards to 0.5 m, away to 1.5 m, towards to 0.4 m. Five walks, turning at each peak and dip.

Questions and analysis

1. Which run shows a **constant speed** the whole way? Run **1**

2. Which run shows someone **at rest** the whole time? Run **3**

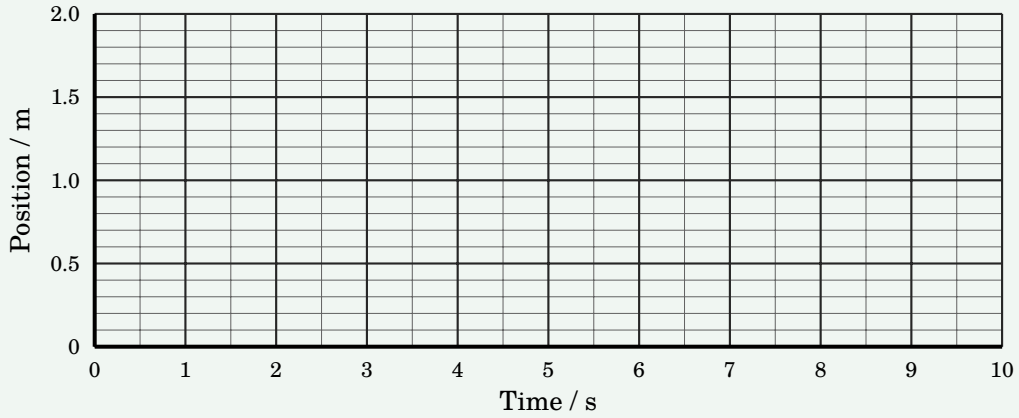
3. Which run shows moving **away** and **speeding up**? Run **4**

4. Which run was the hardest to match? Use your description to explain what made it hard.

Model answer: Run 9, because it has five separate walks and I had to turn round at exactly the right moment each time. Every change of direction is a peak or a dip on the graph, so a late turn shows straight away.

To go further — design your own run

Sketch a target run of your own on these axes: at least three parts, and every part something a person can walk. Swap sheets with another group. Can they match it first time?



Check: no part of your line is vertical, and it never goes below 0.3 m.